DAY 13 | POST-READ

# Testing and Data Engineering: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

## Part A: Summary of today

| Topic | What to remember |
|---|---|
| Testing pyramid | Many unit tests, some integration tests, few end-to-end tests. |
| PyTest basics | test_*.py files, test_* functions, plain assert, pytest -q. |
| pytest.raises | Checks that code raises the expected exception. |
| parametrize | One test, many input and output cases. |
| Fixtures | Reusable setup injected by name, like Depends(). |
| API tests | TestClient + fixtures; check status codes and JSON. |
| React tests | Vitest + React Testing Library: render, find by role or text, click, expect. |
| Data lifecycle | Generate, ingest, store, process, serve. |
| ETL vs ELT | Transform before loading vs load raw then transform in the warehouse. |
| Ingestion | Full, incremental, streaming, API pull. Pipelines must be idempotent. |

## Part B: Going beyond today

### 1. Test coverage

```
pip install pytest-cov
pytest --cov=. --cov-report=term-missing
```

Coverage shows which lines your tests never run. High coverage is a guide, not a goal; a few meaningful tests beat many shallow ones.

### 2. Test-driven development (TDD)

Write a failing test, write just enough code to pass it, then tidy the code. Red, green, refactor. It keeps code small and focused.

### 3. Mocking external calls

```
def test_weather(monkeypatch):
    monkeypatch.setattr("app.fetch_temp", lambda city: 30)
    assert app.advice("Pune") == "Stay hydrated"
```

### 4. Data quality checks

Pipelines need tests too: no null ids, no duplicate keys, amounts within a sensible range, row counts close to yesterday's. Tools such as Great Expectations or dbt tests formalise this.

### 5. Data lakes, warehouses and lakehouses

A lake stores raw files cheaply (any format). A warehouse stores clean tables for fast SQL. A lakehouse combines both, which is what Spark and Delta Lake are often used for; you meet Spark tomorrow.

## Part C: Check yourself

- Why should most tests be unit tests?
- What does a fixture do?
- Why do React Testing Library tests look for text and roles instead of class names?
- Give one advantage of ELT over ETL.
- What does idempotent mean for a pipeline?

## Part D: Practice ideas for tonight

- Reach 10 PyTest tests for your Task Board API, including every error case.
- Add an incremental load to the ETL pipeline that only loads orders newer than the latest date already stored.
- Write data quality tests that fail if any order has a negative amount.

**Coming up on Day 14:** batch data pipelines and Apache Spark: DataFrames, transformations and joins.

*Otsukaresama deshita (お疲れ様でした): thank you for your hard work today*